

Ablation summary from invariance CV

Mean coefficient of variation across the 48 hyperparameter settings; compared to the C. elegans baseline.

Ablation	Result on each metric (mean CV)	Inc/dec/same vs C. elegans	Interpretation
C. elegans	MC 0.171 IPC 0.276 KR 0.205 GR 0.275	baseline	Low-dispersion reference: biological topology, sign balance, and heterogeneous weights act together.
CE Gaussian wt.	MC 0.293 IPC 0.198 KR 0.478 GR 0.524	MC inc +72%; IPC dec -28% KR inc +134%; GR inc +90%	Preserving topology alone is not enough; dropping biological signs and magnitudes moves MC/KR/GR into a higher-CV regime.
ER Gaussian wt.	MC 0.350 IPC 0.219 KR 0.571 GR 0.580	MC inc +105%; IPC dec -21% KR inc +179%; GR inc +111%	Random topology plus Gaussian weights loses the low-CV profile, showing that connectome structure still matters.
WS Gaussian wt.	MC 0.346 IPC 0.218 KR 0.581 GR 0.584	MC inc +103%; IPC dec -21% KR inc +184%; GR inc +112%	Matches ER across metrics in the thesis; small-world organization alone does not recover invariance.
Sign-pres. Gaussian	MC 0.168 IPC 0.287 KR 0.240 GR 0.324	MC same -1%; IPC same +4% KR inc +17%; GR inc +18%	With topology and signs fixed, random magnitudes behave like the uniform control; MC is closest to the connectome.
Sign-pres. uniform	MC 0.167 IPC 0.289 KR 0.233 GR 0.318	MC same -2%; IPC inc +5% KR inc +14%; GR inc +15%	Uniform and Gaussian sign-preserving magnitudes are similar, but neither fully reproduces IPC/KR/GR.
Sign-pres. +/-1 wt.	MC 0.163 IPC 0.288 KR 0.217 GR 0.299	MC dec -5%; IPC same +4% KR inc +6%; GR inc +9%	Collapsing weights to +/-1 preserves much of IPC/KR/GR, but thesis equivalence tests say MC is not recovered.
+/-1 sign shuffle	MC 0.164 IPC 0.289 KR 0.226 GR 0.306	MC same -4%; IPC inc +5% KR inc +10%; GR inc +11%	Preserving only global E/I balance is close to local +/-1; exact edge-by-edge sign placement is less important for IPC/KR/GR.
Unsigned binary wt.	MC 0.161 IPC 0.289 KR 0.213 GR 0.290	MC dec -6%; IPC inc +5% KR same +4%; GR inc +5%	Low CV alone is not the full biological profile; this unsigned model was equivalent to the connectome only for IPC.

Higher CV = less invariant; lower CV = more invariant. Inc/dec/same uses a +/-5% mean-CV threshold.